2017_3(a)

Integrity Constraint: set of rules that are applied to the relations and relationships to ensure that the overall validity, integrity, consistency of the data is maintained.

1. Domain Integrity Constraint: set of rules that restrict the kinds of values a column hold in a table.

2. ~~Entity Integrity~~ Key Constraint: used to ensure the uniqueness of ~~each record~~ a column. in a table. restriction on duplicate values. (unique, primary key, not null, check) (used to identify row.)

3. Referential Integrity Constraint: ensures the relationships between ~~two~~ tables are valid and that foreign key values in one table match primary key values in another table/null.

Entity Integrity

1. Key Constraint:

→ each row uniquely identifiable through primary key.

→ guarantees no primary key is null/duplicated.

Key → primary key, foreign key, unique ensure the uniqueness and non null property of a specific set of columns.

Entity Integrity:

ensure the validity and uniqueness of primary key of a table.

2017-3(b)

Trigger is a special type of stored procedure that is automatically executed in response to certain database events. The purpose of a trigger is to perform specific action(s) based on the occurance of the triggering event.

```
create trigger x
~~on delete~~
after delete on accounts
for each row
declare
    cursor c is select account_no from accounts
                                    join depositor using
                                    (account_no)
where
```

account(a_no, balance) ✓

depositor(c_id, ~~c_name~~, a_no)

```
create trigger kill
after delete on account
for each row
declare
    cnt number;
begin
    select count(*) into cnt from depositor where
c_id = (select c_id from depositor where
 a_no = :old.a_no);
    if cnt = 0 then
        delete from depositor where c_id=
(select c_id from depositor where a_no = :old.a_no);
    endif;
end;
```

2017_4(a)

Attribute closure α^+ is defined as the set of attributes that are functionally determined by α* under F.

Uses:

(i) Testing superkey:

$R \subseteq \alpha^+$ α – superkey

(ii) Testing functional dependency:

$\alpha \rightarrow \beta$

$\beta \subseteq \alpha^+ \longrightarrow (\alpha \rightarrow \beta)$ holds

(iii) Computing F^+:

For each $\gamma \subseteq R$ find γ^+

for each $\delta \subseteq \gamma^+$, $\boxed{\gamma \rightarrow \delta}$

2017_4(b)

BCNF — $\alpha \rightarrow \beta$ (i) trivial (ii) α superkey

3NF — $\alpha \rightarrow \beta$ (i) trivial (ii) α superkey (iii) $\beta - \alpha$ contained in candidate key of R.

each attribute A in

3NF > BCNF → preserve dependency by allowing some redundant attributes (both lossless decomposition).

Book(title, author_name, author_designation, type, price, publisher).

type: ✗ publisher: ✗

(✗) title → publisher, type violets (i), (ii)

R_1(title, publisher, type) Book(title, author_name, author_designation, price).

BCNF

(✗) type → price (✗)

(✗) author_name → author_designation. (✗) (i), (ii)

$R_2 = (title, author\_name, price)$

BCNF

$R_3 = (author\_name, author\_designation)$

BCNF.

2017_4(c)

Structured data:

→ highly organized and follows a predefined data model / schema → elements are addressable for effective analysis.

→ based on relational database table.

→ less flexible

Student

ID	Name
10	Nihal

Semi structured data:

doesn't have fixed schema but possesses some level of organization that make it easier to analyze.

→ based on XML/RDF/JSON

→ more flexible than structured data

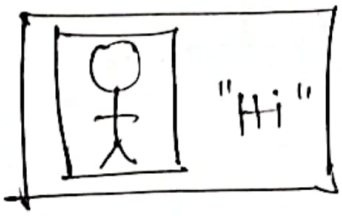
```
{"name": "Nihal",
  "address": {
      "street": "123",
      "city" : "ABC"
  },
  "phone" : [
     { "type" : "home",
       "number" : "123"
     }
     {  ____
        ____
     }
],
                    "hobbies" : ["__","__","__
                }
```

Unstructured data:

- data with no predefined organizational form / specific format.

→ based on character / binary data

→ flexible



2017_2(d)

	max	min
no. of pointers in leaf / non leaf node	n	$\lceil \frac{n}{2} \rceil$
no. of values in leaf/non leaf node	n-1	$\lceil \frac{n-1}{2} \rceil$

x — keys

$$x \times 9 + (x+1) \times 7 \leq 512$$

$$\Rightarrow 9x + 7x + 1 \leq 512$$

$$\Rightarrow 16x \leq 511$$

$$\Rightarrow x \leq 31.93$$

$$x \simeq 31. \quad (A).$$